14 aprile

P' rilassato (RL) → $\underline{x}^\circ$
P intero → $\underline{x}^*$

$P \subset P'$

$\max \underbrace{c^T x^\circ}_{\text{upper bound}} \geq c^T x^*$

$c^T x^\circ < c^T x^*$
$x^\circ \in P' \quad P \subset P'$
$\underline{x}^* \in P$

$x_B = B^{-1} \underline{b}$
$A \ \underline{b}$
$\frac{1}{\det(B)} \ adj(B)$

$Q_1 = \{\} \quad Q_2 = \{\}$

$$\begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

NON TOTALMENTE UNIMODULARE

1) Quando processo la prima colonna $Q_1 = \{1\}, Q_2 \{2\}$
2) $Q_1 = \{1\}, Q_2 = \{2, 3\}$
3) fallisce

altro esempio

$$\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

1) $Q_1 = \{1\}, Q_2 = \{2\}$
2) $Q_1 = \{1\}, Q_2 = \{2, 3\}$
3) $Q_1 = \{1, 4\}, Q_2 = \{2, 3\}$
4) $Q_1 = \{1, 4\}, Q_2 = \{2, 4\}$ → TOTALMENTE UNIMODULARE

altro esempio

$$\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 0 & 1 & -1 & 0 \\ -1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

1 → $Q_1 = \{1, 2\}, Q_2 = \emptyset$
2 → $Q_2 = \{1, 2, 3\}, Q_2 = \emptyset$
3 → $Q_3 = \{1, 2, 3\}, Q_2 = \emptyset$
4 → $Q_4 = \{1, 2, 3, 4\}, Q_2 = \emptyset$
5 → $Q_5 = \{1, 2, 3, 4, 5\}, Q_2 = \emptyset$

Cut

$x \leq 2{,}7$
↓
$x \leq 2$

x intera $\quad \forall^m$
$x \leq b \quad b \in \mathbb{Q} \setminus \mathbb{Z}$
$b = \lfloor b \rfloor + f$
$x \leq \lfloor b \rfloor + f \quad 0 < f < 1$
⇓
$x \leq \lfloor b \rfloor$

$A \underline{x} = \underline{b}$
$m \times n$
$u = \begin{bmatrix} u_1 \\ \vdots \\ u_m \end{bmatrix}$

$\lfloor u^T A \rfloor \underline{x} \leq \underline{u}^T \underline{b}$
$\leq \lfloor \underline{u}^T \underline{b} \rfloor$

$A\underline{x} = \underline{b}$

$\frac{1}{2} \quad x_1 + x_2 + s_1 = 1$
$\frac{1}{2} \quad x_2 + x_3 + s_2 = 1$
$\frac{1}{2} \quad x_1 + x_3 + s_3 = 1$

$\underline{u}^T A \underline{x} = \underline{u}^T \underline{b}$

$x_1 + x_2 + x_3 \leq 1$ È ANCHE UN TAGLIO

$x_1 + x_2 + x_3 + \frac{1}{2} s_1 + \frac{1}{2} s_2 + \frac{1}{2} s_3 = \frac{3}{2}$

Costanti

$x_{Bi} + \sum_{j \in R} y_{ij} x_j = y_{i0}$

R = {insieme degli indici delle variabili fuori base}

$y_{i0} = \lfloor y_{i0} \rfloor + f_{i0}$ $\quad 0 < f_{i0} < 1$
non è intero

TAGLIO DI GOMORY IN FORMA INTERA

$x_{B_i} + \sum_{j \in R} \lfloor y_{ij} \rfloor x_j + \underbrace{\sum_{j \in R} f_{ij} x_j}_{\geq 0} = y_{i0}$

$x_{B_i} + \sum_{j \in R} \lfloor y_{ij} \rfloor x_j \leq y_{i0} = \lfloor y_{i0} \rfloor + f_{i0}$

x_B / x_{Bi}		x_N / x_B	
x_{Bi} i	1	y_{ij}	y_{i0}

$\underline{u}^T = (B^{-1})^i$ LA RIGA i-MA DI B^{-1}

$$x_{Bi} + \sum_J y_{ij} x_J = y_{i0}$$

$$x_{Bi} = y_{i0} - \sum_{J \in R} y_{ij} x_J$$

$$y_{i0} - \sum_{J \in R} y_{ij} x_J + \sum_{J \in R} \lfloor y_{ij} \rfloor x_J \leq \lfloor y_{i0} \rfloor$$

$$\lfloor y_{i0} \rfloor + f_{i0} - \sum_{J \in R} \lfloor y_{ij} \rfloor - \sum_{J \in R} f_{ij} x_J + \sum_{J \in R} \lfloor y_{ij} \rfloor x_J \leq \lfloor y_{i0} \rfloor$$

$$\Longrightarrow \sum_{J \in R} f_{ij} x_J \geq f_{i0}$$

TAGLIO DI GOMERY IN FORMA FRAZIONARIA